
Build the Geometry In, Don’t Hope It Emerges: A Reflection-Symmetry Audit of a Skeleton World Model

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Abstract

1 We describe a registered protocol for testing whether a self-supervised skeleton
2 model learns useful left–right structure. The target is a coordinate-derived com-
3 parison of motion on the two sides, not a diagnosis or clinical gait measure. The
4 quality-controlled cohort contains 625 pose sequences from 93 source videos.
5 Five outer folds hold source videos out of both representation training and read-
6 out fitting, while four inner folds select the read-out penalty using outer-training
7 sources only. The design supports a narrow held-out-video claim and cannot es-
8 tablish unseen-person or clinical generalization. The complete paper run and gov-
9 ernance reviews are pending, so no v2.1 empirical conclusion is reported.

10 1 Motivation and target

11 A human pose representation has a natural left–right structure. If a skeleton is reflected horizontally
12 and its anatomical left and right landmarks are exchanged, a signed left-versus-right motion quantity
13 should reverse sign. We use this fact to test a self-supervised Skeleton Joint-Embedding Predictive
14 Architecture (S-JEPA). The model learns by predicting hidden latent content; it does not receive the
15 laterality target or dataset annotation during representation training.

16 For the left and right shoulders, knees, ankles, heels, and foot-index points, the target compares
17 median speeds only on frame transitions where both sides are observed at both endpoints. For pair
18 k ,

$$c_k = \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + \epsilon}, \quad y = \text{mean}_k c_k.$$

19 All five pairs must each provide at least eight shared valid transitions. Mirroring must negate y ,
20 changing invalid coordinate placeholders must leave it unchanged, and mirroring twice must restore
21 the input. Interpolated values may enter encoder input but never define the target.

22 2 Cohort and source-level split

23 The Gait Abnormality in Video Dataset (GAVD) inventory contains 666 annotation files for 103
24 source videos [Ranjan et al., 2025, GAVD project, 2026]. A matching pose archive is available
25 for 642 annotations. Pre-specified quality-control rules retain 625 pose sequences from 93 source
26 videos and exclude 17 sequences. Inclusion depends on target computability and pose coverage,
27 never on the sign or magnitude of the target.

Table 1: Five registered outer folds.

Fold	Train sources	Test sources	Train sequences	Test sequences
0	74	19	436	189
1	74	19	443	182
2	74	19	553	72
3	75	18	548	77
4	75	18	520	105

The split is created at the source-video level because one source can yield several related sequences. Every sequence, mirror, and augmentation inherits its source’s fold, preventing material from a test video from entering the corresponding training set.

Sequence counts vary because videos contribute different numbers of sequences. Every source video is tested exactly once. GAVD does not supply persistent person identifiers, so this is held-out-video evaluation, not held-out-person evaluation.

3 Training, validation, and evaluation

A fresh encoder is trained for every outer fold, seed, and variant. The registered paper design contains five folds, five optimization seeds, and two variants—vanilla and reflection-augmented—for 50 encoders. Each encoder uses all of its outer-training sources and follows a fixed 300-epoch schedule. It has no early-stopping validation set. Sampling chooses a source uniformly before choosing one of its sequences, preventing videos with many sequences from dominating.

Validation is reserved for the linear ridge read-out. Four inner folds are built entirely inside each outer-training set. In one inner round, 55–57 sources fit a candidate read-out and 18–19 sources validate its ridge penalty. After selection, the read-out is refitted on all 74–75 outer-training sources and evaluated once on the untouched 18–19 outer-test sources. Fitted preprocessing uses outer-training data only.

The inner fitting partitions contain 266–450 sequences, and the rotating validation partitions contain 80–197 sequences. Their sizes vary because sources contribute different numbers of sequences. None includes an outer-test source.

The primary measure is source-balanced R^2 , which gives every source video equal total weight. Comparison lanes include a paired untrained encoder, a target-component self-consistency oracle, measured nuisance features, analytically antisymmetric read-outs, and direct encoder-token reflection tests. The oracle verifies target reconstruction and is not a learned baseline. Exact sign change from a constructed wrapper proves a mathematical property of the wrapper; evidence that training learned useful symmetry additionally requires positive held-out prediction and improvement over paired initialization.

Five seeds describe optimization variation. Separately, 2,000 source-video bootstrap resamples describe uncertainty associated with the observed source collection while keeping a video’s sequences together. These intervals are conditional on the fitted cross-validation pipeline and do not imply population or unseen-person generalization.

4 Claim boundary, ethics, and status

The strongest possible claim is post-development, within-GAVD cross-validated performance on held-out source videos. The protocol does not support diagnosis, clinical validity, prevalence, treatment effects, or generalization to unidentified people in new videos. Folder names are dataset annotations, not diagnoses.

This document describes laterality v2.1. Earlier transductive estimates are superseded and are not results from this design. No v2.1 empirical conclusion will be reported until all registered folds, seeds, variants, evaluations, and lineage checks are complete. Submission and artifact release also remain blocked until the institutional ethics determination, data-use review, and derived-pose release review contain genuine dated internal references. No raw video or identity-bearing frame is redistributed, and derived artifacts remain non-public unless completed reviews permit release.

70 **References**

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